

AILENI VARSHITH REDDY

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Professional Summary

Passionate software engineer with 1+ year of hands-on experience building and shipping production-ready software systems. Strong ability to solve complex problems through systematic design and collaborative development. Proficient in applying core computer science fundamentals, including data structures, algorithms, and system design, to develop reliable and scalable solutions. Experienced in writing clean, maintainable code and following testing best practices. Highly motivated to grow technical expertise and contribute effectively within a world-class engineering team.

Education

Vidya Jyothi Institute of Technology (VJIT)
B.Tech in Information Technology, CGPA: 8.44/10

Expected Jun 2026
Hyderabad, Telangana

Professional Experience

Glow Tech Solutions

Software Engineer (Full-Stack Development Intern)

Dec 2024 - Mar 2025
Hyderabad, Telangana

- Architected and deployed scalable backend services using **Node.js, Express.js, and RESTful APIs** handling **500+ daily active users**, optimized MongoDB queries using indexing and aggregation pipelines reducing latency by **45%**.
- Developed responsive **React and Next.js** frontend applications with **TypeScript**, improving performance by **40%** through code splitting, lazy loading, and CSS optimizations achieving **95+ Lighthouse scores**.
- Designed and tested **REST APIs** using Postman, implemented comprehensive error handling, input validation, and middleware, collaborated with backend team on API contracts.
- Established **CI/CD pipelines** using GitHub Actions and Docker, reducing deployment time by **30%**, implemented unit and integration tests, participated in code reviews.
- Shipped 8+ features across 3 releases using Agile methodology in fast-paced startup environment.

Intrinsic Science Labs

Software Engineer (Full-Stack Development Intern)

Sep 2024 - Dec 2024
Hyderabad, Telangana

- Developed reusable **React components** using modern patterns and hooks, optimized component rendering; reduced development time by **25%** across 3+ projects.
- Achieved **90%+ code coverage** with **Jest and React Testing Library**, wrote comprehensive unit, integration, and end-to-end tests ensuring code reliability.
- Implemented state management with **Redux**, debugged complex state flow issues, integrated RESTful APIs with proper error handling and loading states.
- Built backend services using **NestJS**, designed modular architecture with controllers, services, and middleware, implemented dependency injection for scalability.
- Collaborated with design and backend teams, conducted design and code reviews, mentored peers on TypeScript and React best practices.

Key Projects

InboxSync – AI-Powered Email Management SaaS | *Node.js, TypeScript, React, PostgreSQL, pgvector, Elasticsearch, OpenAI, Railway, Vercel* | [GitHub](#)

- Full-stack production SaaS platform enabling real-time multi-account email management with **sub-1-second IMAP sync** using IDLE mode instead of polling, processing **10,000+ emails daily**.
- Implemented **RAG (Retrieval-Augmented Generation)** pipeline: stored training emails as pgvector embeddings, performed vector similarity search to retrieve context, generated AI-powered personalized email replies achieving **85% suggestion acceptance rate**.
- Backend architecture: Multi-tenant system with OAuth authentication (Gmail/Outlook), real-time email ingestion, automatic categorization using GPT-4o-mini into 5 business categories with **92% accuracy**.
- Advanced search infrastructure: Elasticsearch indexing enabling sub-50ms full-text queries across sender, subject, body; implemented efficient pagination and filtering for 10K+ email datasets.
- Production operations: Deployed on Railway (backend) + Vercel (frontend), achieved **96%+ uptime**, integrated Sentry for error tracking, implemented structured logging, automated CI/CD with GitHub Actions.
- Performance optimization: Database query optimization reducing latency by 40%, pgvector similarity search tuning, API response caching reducing redundant LLM calls by 60%.

Syntra AI Agentic Assistant – LLM-Powered Autonomous Platform | *Next.js, TypeScript, LangChain, IBM Watson NLP, Node.js, Clerk Auth, Vercel* | [GitHub](#)

- Full-stack system design integrating large language models (LLMs) into production web platform, handled **500+ autonomous workflows/month** with sub-200ms response latency.
- Backend architecture: Built Node.js Monolithic using LangChain for LLM orchestration, implemented agentic task execution framework allowing autonomous agents to complete multi-step workflows, integrated IBM Watson NLP for natural language understanding.
- Frontend-to-Backend integration: Designed serverless Next.js architecture with Clerk authentication and RBAC, built intuitive UI for workflow management, monitoring, and agent interaction, implemented real-time status updates using WebSockets.
- Performance & reliability: Optimized inference latency through prompt caching and result memoization, implemented exponential backoff retry logic for LLM API calls, deployed on Vercel with CI/CD pipelines and error tracking via Sentry.
- Problem-solving approach: Abstracted complex LLM integration patterns into reusable modules, designed database schema for storing workflow history, execution logs, and agent state for debugging and optimization.

Technical Skills

- Programming Languages:** JavaScript, TypeScript, Java, Python, SQL, HTML/CSS
- Frontend:** React, Next.js, Redux, Context API, Tailwind CSS, Material-UI, Responsive Design
- Backend:** Node.js, Express.js, NestJS, REST API Design, Middleware, Authentication (JWT/OAuth)
- AI/Systems:** LangChain, LLM Integration, LangGraph, OpenAI API, Prompt Engineering, RAG, pgvector

- **Databases:** PostgreSQL, MongoDB, Prisma ORM, Query Optimization, Vector Embeddings, Elasticsearch
- **Testing:** Jest, React Testing Library, Unit Testing, Integration Testing, API Testing (Postman)
- **DevOps/Deployment:** Git, GitHub Actions, CI/CD, Docker, Vercel, Railway, AWS (EC2, S3), Linux
- **System Design:** Scalable Architecture, Real-time Systems, Microservices, Distributed Systems, Caching, Load Balancing
- **Soft Skills:** Problem-Solving, Communication, Cross-functional Collaboration, Code Review, Agile/Scrum

Achievements & Recognition

- **Smart India Hackathon 2025 Finalist** – Top 5000+ teams nationally, demonstrated system design, ML integration, and real-world problem-solving for government use case.
- **Full-Stack Production Deployment** – Built and shipped InboxSync SaaS with 96%+ uptime, processing 10,000+ emails daily with sub-50ms search latency.
- **Scalability Expert** – Built systems serving 1,000+ concurrent users, optimized performance (40% frontend, 45% database, 35% ML inference, 60% API caching) across multiple projects.
- **Code Quality & Testing** – Maintained 90%+ test coverage, led code review initiatives emphasizing clean architecture and maintainability.
- **Technical Leadership** – Lead TEDXVJIT technical team for 2025, mentored junior developers on full-stack development and best practices.